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Palaeomagnetism of the Palaeoproterozoic ultramafic intrusion near Lake Konchozero, Southern Karelia, Russia

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Abstract

New palaeomagnetic results are presented from the recently dated Palaeoproterozoic ultramafic Konchozero sill, and associated basalts (three sites, 38 oriented samples). Three stable components of remanence have been isolated during thermal and alternating field demagnetisation. The component I, with a mean direction of $D=103^\circ$, $I=40^\circ$, $k=18$, $\alpha_{95}=11^\circ$ ($N=11$ samples), pole position of 14°S, 282°E, has been obtained from the unaltered deeper part of the sill and from baked schists. The study of the baked contact confirms the conclusion that component I is supposed to be primary and corresponds to the Sm–Nd age of the sill of 1974 ± 27 Ma. The palaeopole of component I is not consistent with the accepted Fennoscandian apparent polar wander path (APWP) for the period 2120–1880 Ma, and for that part the Fennoscandian APWP should be revised. Two other components (component II: $D=349^\circ$, $I=39^\circ$, $k=35$, $\alpha_{95}=6^\circ$, $N=19$ samples, pole position 49°N, 231°E; and component III: $D=17^\circ$, $I=41^\circ$, $k=44$, $\alpha_{95}=5^\circ$, $N=19$ samples, pole position 50°N, 190°E) fit the APWP well, with palaeomagnetically estimated ages of ca. 1860 and 1760 Ma respectively. © 1999 Elsevier Science B.V. All rights reserved.

Keywords: Fennoscandia; Palaeomagnetism; Palaeoproterozoic; Russia

1. Introduction

While analysing the drift history of the Fennoscandian shield in the Archaean and Palaeoproterozoic based on the apparent polar wander path (APWP) of Elming et al. (1993), it is obvious that data are sparse between ca. 2400 and 1900 Ma. The main reason is lack of reliable palaeomagnetic data from well-dated rocks of these ages. Only one 'key' pole with the age of 2120 Ma was calculated for this period. This pole is based on only two grade C [according to the Briden and Duff (1981) scale modified by Elming et al. (1993)] palaeomagnetic determinations. One of these determinations was obtained by Gooskova and Krasnova (1985) from the Muezero gabbro-diabase dyke, Karelia. Another one, published by Mertanen et al. (1989), is the secondary component E from early Proterozoic layered intrusions, northern Finland. These two poles are located about 50° apart.

The above-mentioned interval of the Palaeoproterozoic is very important for under-

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standing the tectonic history of Fennoscandia. At ca. 2000 Ma the Ukrainian shield was probably attached to the Fennoscandian shield (Bogdanova et al., 1994).

According to Zonenshain et al. (1990), the oldest part of Fennoscandia was divided into four separate blocks [Fig. 1(a)]: the Karelian (together with a part of Voronezh massif and a large part of the Ukrainian shield), Belomorian, Kola and Murmansk blocks. These blocks probably collided and formed the Fennoscandian shield between 2000 and 1700 Ma. Recent palaeomagnetic data (Khramov et al., 1997) suggest that for the Karelian, Belomorian and Kola blocks it is possible to draw separate APW paths before 2000 Ma.

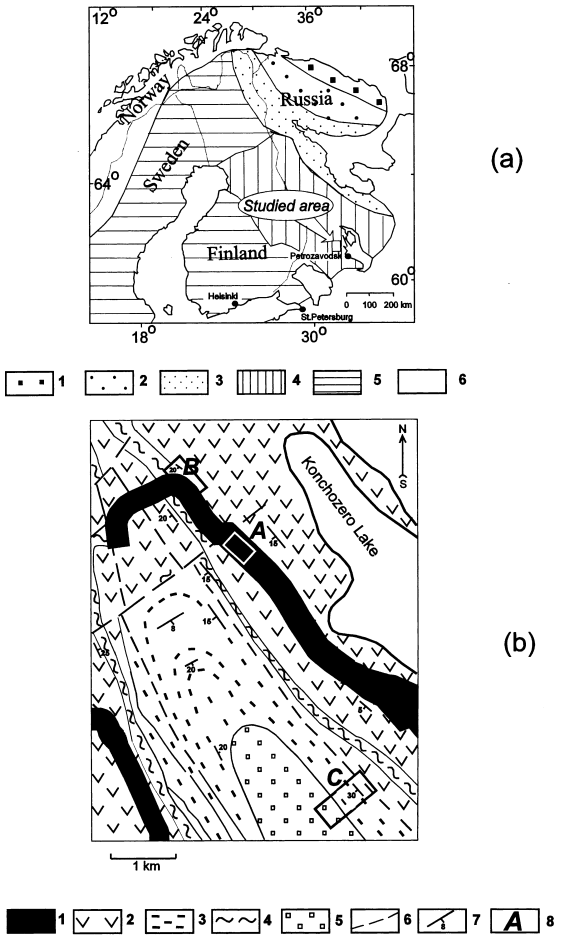


Fig. 1. (a) Tectonic sketch map of the Fennoscandian shield. 1=Murmansk block; 2=Kola block; 3=Belomorian block; 4=Karelian block; 5=Svecofennian block; 6=Phanerozoic cover.(b) Geological map of the sampled area. 1=Konchozero peridotite sill; 2=diabases interbedded with tuffs and sedimentary rocks; 3=olivine and picritic basalts interbedded with tuffs and tuffites; 4=tuffs, tuffites and siliceous schists; 5=pyroxene-olivine basalts; 6=faults; 7=bedding; 8=sampled sites (see text).

Puchtel et al. (1995) obtained an Sm–Nd age of 1974±27 Ma for the Konchozero sill near Petrozavodsk city, Karelia, Russia. The petrographic and geochemical studies show that the process of cooling and crystallisation took place long enough for the isotopic equilibrium to be achieved before the closure of the Sm–Nd system inside the individual mineral grains. It means that the Sm–Nd isochron age of 1974±27 Ma corresponds to the time of the intrusion. The abundance of mineralogical, petrological and other studies of this intrusion, the presence of an exposed contact with country schists (Kulikov et al., 1987), together with the new age determination, led us to conduct a palaeomagnetic study of the sill and its country rocks. Determination of a new palaeomagnetic pole position for the poorly studied early-middle Palaeoproterozoic time was the main aim of our study.

2. Geology and sampling

According to Sokolov et al. (1987), lower Proterozoic (Karelian) strata in Southern Karelia can be subdivided into six lithostratigraphic units: Sumian, Sariolian, Jatulian, Ludikovian, Livvian and Vepsian Groups. The Ludikovian Group (ca. 2100–1950 Ma) consists of two sub-units: Zaonezhie and Suisarian. The thickness of the Ludikovian sequence is about 1700 m. Both sedimentary and volcanic rocks are present.

The sampled area (Fig. 1) is situated near Lake Konchozero (about 40 km NW of Petrozavodsk). Several outcrops of Suisarian volcanic and sedimentary rocks and the body of the Konchozero peridotitic sill are situated here. The sill was probably a magmatic chamber for the Suisarian basalt lavas (Puchtel et al., 1995). Its thickness varies from 200 m in the central part to 10 m distally. The intrusion is characterised by the presence of chilled zones and an asymmetric internal structure.

The bottom part of the sill (some 40–80 m thick) consists of olivine and pyroxene cumulates with the abundance and grain size of olivine decreasing towards the upper part of the sill. In the uppermost part, about 20 m thick, the peridotites are replaced by gabbro. The sill and country rocks together form a gentle syncline. It is possible that this structure was formed during the Svecofennian tectonic epoch (ca. 1870 – 1860 Ma). However it is also possible that the body was intruded into originally inclined beds in the depression of the basin. The studied contact zone [Fig. 1(b), site B] consists of the chilled zone (about 4 m) and the zone of picrites (10–15 m).

The sequence of Suisarian picritic basalts and tuffaceous schists overlies the intrusion. These basalt flows are pale green-grey rocks, containing tabular and prismatic clinopyroxene phenocrysts and subordinate serpentine pseudomorphs after platy olivine phenocrysts, set in a fine-grained matrix. Normally the flows are rather uniform in structure and vary in thickness by several metres to several tens of metres. The total thickness of the section exceeds 420 m.

The sill, schists and basalts were subjected to low-grade metamorphism (greenschist facies). Analysis of polished sections shows the presence of titanomagnetite, both primary and secondary magnetite and haematite in all studied parts of the sill. However, in the lower part of the sill, the peridotites show little or no effects of metamorphism.

Three localities have been chosen for sampling: the bottom, central and upper parts of the body at site A [Fig. 1(b)], its central part, Suisarian schists and their baked contact at site B and overlying Suisarian picritic basalts and tuffaceous schists at site C. Site A is close to the sampling locality of Puchtel et al. (1995) for the Sm–Nd age. Thirty-eight large oriented blocks were sampled: 7 blocks from peridotites of the bottom part of the sill (site A, samples 1–7); 12 blocks from peridotites, picrites and gabbro of the central and upper parts of the sill (site A, samples 8–13; site B, samples 19–24); 5 blocks from baked schists (site B, samples 16–18, 25, 26); 4 blocks from unbaked schists (site B, samples 14, 15; site C, samples 27, 31); and 10 blocks from picritic basalts (site C, samples 28–30, 32–38). A magnetic compass was used for orientation.

3. Measuring techniques

Specimens (4–12) with a diameter of 2.54 cm and 2.2 cm high were drilled from each oriented block. The natural remanent magnetisation (NRM) was measured with JR-4 and JR-5 spinner magnetometers (Geofyzika, Brno, Czech Republic). Stepwise alternating field (AF) and thermal demagnetisation techniques were used to isolate the characteristic remanent magnetisation (ChRM). AF-demagnetisation was carried out in steps up to 100 mT using the tumbling demagnetiser constructed in the Lund laboratory, Sweden (residual field less than 20 nT). During thermal demagnetisation, changes in magnetic mineralogy during heating were monitored by measuring magnetic susceptibility. The heating was done in a Schonstedt TSD-1 thermal demagnetiser in a field less than 20 nT during heating and cooling.

One pilot specimen from each sample was heated to temperatures of 100, 200, 250, 300, 350, 400, 450, 500, 530, 560, 570, 575, 580, 585, 590, 595 and 600°C, and one specimen was AF demagnetised at 10, 20, 30, 40, 50, 60, 70, 80, 90 and 100 mT. Some specimens possessed a strong magnetisation even after heating to 600°C. Eight specimens were also studied in the Palaeomagnetic laboratory of the University of Florida (Gainesville, FL, USA) using the 2G SQUID magnetometer. Stepwise thermal demagnetisation up to 680°C with 10° steps after 350°C was carried out for those specimens. These additional results gave us only the confirmation of previous data obtained in the laboratory at Lund University.

The results obtained from the pilot collection were used for determining the demagnetisation intervals and the number of steps to be used for the remaining specimens. Depending on their behaviour at critical temperatures and fields they were treated with smaller steps. Typical orthogonal diagrams (Zijderveld, 1967) for each type of rocks are presented in Fig. 5.

Curie temperatures (T_C) were determined from changes in magnetic susceptibility during heating

and cooling in air. A CS2 furnace device coupled to a KLY-2 Kappabridge (Geofyzika, Brno) was used.

Saturation isothermal remanent magnetisation (SIRM) was determined by inducing fields with a Molspin magnetic pulse charger for fields up to 200 mT and a Redcliffe 700 BSM pulse charger for fields up to 4 T. Further, the AF and thermal demagnetisations gave information on the magnetic properties of the NRM carriers.

4. Magnetic properties

Initial measurements of NRM and magnetic susceptibility have shown that the studied rocks can be divided into three different groups (Fig. 2). Peridotites from the bottom part of the sill (site A) have NRM ranging from 1.1 to 44.0 A m^{-1}, and magnetic susceptibility from 34.5 to 97.8×10^{-3} SI units. Koenigsberger's Q ratios are between 0.7 and 27.2, indicating that these rocks have stable remanences (Piper, 1987). The small group of peridotites and picrites from the central part of the sill (sites A and B) show 'intermediate' magnetic properties with NRM intensities ranging from 36 to 58 mA m^{-1} and susceptibility from 2.2 to 2.8×10^{-3} SI units. Their Koenigsberger's Q ratios are between 0.39 and 0.57. All palaeomagnetic directions retrieved from this intermediate group are similar to the results from the upper part of the sill (see below). Picrites and gabbros of the upper part of the sill, country picritic basalts and schists (sites A, B and C) are weakly magnetic: NRM ranges from 0.8 to 10.1 mA m^{-1} and magnetic susceptibility from 0.7 to 1.1×10^{-3} SI units. Koenigsberger's Q ratios are between 0.02 and 0.29.

The low field susceptibility versus temperature curves for specimens from the various rock types are presented in Fig. 3. There are two types of curves. The first type is characteristic for the peridotites of the deep part of the sill [Fig. 3(a)]. They show small differences between heating and cooling curves and point to a very narrow Curie temperature range close to that of pure magnetite. The study of polish sections show the presence of both primary and secondary magnetite in these peridotite together with some titanomagnetite and haematite. Secondary magnetic minerals were created partly during the process of serpentinisation. This process could occur during the cooling of the intrusion at ca. 600°C. It is possible that the

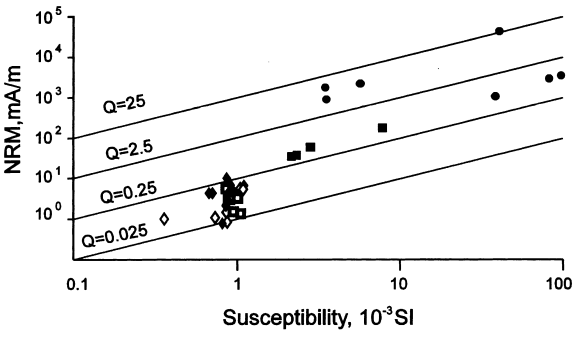


Fig. 2. Remanence-susceptibility relations of different rock types. Closed circles represent peridotites from the bottom part; closed squares, peridotites and picrites of the central part; open squares, picrites and gabbros of the upper part; closed diamonds, picritic basalts; open diamonds, schists.

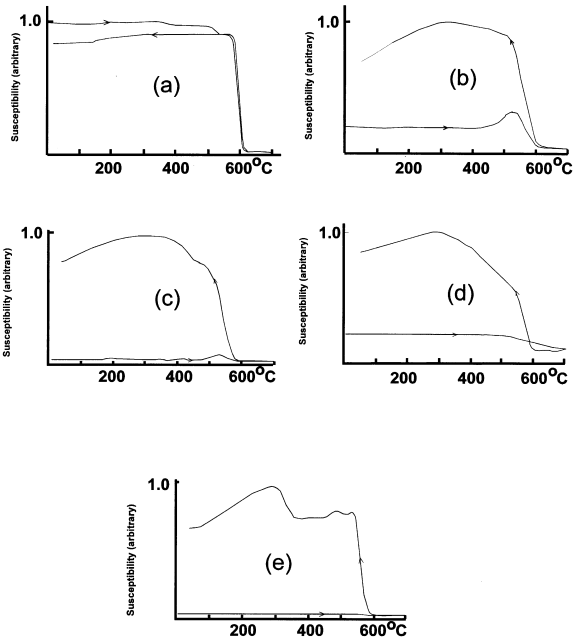


Fig. 3. Examples of thermomagnetic curves: (a) peridotite, lower part of the sill (site A); (b) peridotite, central part of the sill (site A); (c) baked schist (site B); (d) country (non-baked) schist (site C); (e) picritic basalt (site C).

serpentinisation also took place later, during the Svecofennian folding (after ca. 1880 Ma). All other studied rocks [central and top parts of the sill, baked and non-baked country schists, country picritic basalts shown in Fig. 3(b)–(e)] show differences in detail but mainly similar behaviour for their heating and cooling curves. During heating their susceptibility varies little with temperature until close to the range of Curie temperatures for the titanium-poor titanomagnetites, or close to the Hopkinson peak. A sharp increase in susceptibility occurs during cooling which is probably caused by the generation of magnetite from the non-magnetic matrix. There are many possible explanations for such behaviour, but, most important, it seems that these rocks were probably not subjected to intense heating after reaching their present magnetic state. The behaviour of magnetic susceptibility measured after cooling from each temperature step during thermal demagnetisation confirms such a suggestion: it changes significantly only at temperatures higher than 550°C. The mineralogical analysis of these rocks shows the predominance of metamorphic secondary magnetite and haematite.

There are also two types of SIRM curves (Fig. 4). Multi-domain magnetite is mostly responsible for the magnetic properties of the peridotite from the lower part of the sill [Fig. 4(a)]: IRM grows quickly in relatively low magnetic fields, and the coercivity of remanence is small. However, IRM continues to grow even in fields of about 4 T, which is evidence for the presence of haematite. Thermal demagnetisation of the peridotite NRM confirms this conclusion (see Section 5). SIRM curves for the peridotite from the central part of the sill and all other studied rock types [Fig. 4(b)–(d)] point to haematite as the dominant magnetic mineral: remanence saturation was not achieved in a field of 4 T and the coercivity is high.

The mineralogical study of thin and polished sections shows that the metamorphic changes in the peridotite from the lower part of the sill were much weaker than those for other studied rocks.

5. Palaeomagnetic analysis

The characteristic components of NRM were isolated using a least-square algorithm (Torsvik, 1986), combined with analysis of stereograms. Fisher (1953) statistics were used for calculations. Mean results based on sample data (mean results from the studied specimens per sample) are given in Table 1.

Three components of magnetisation were isolated during thermal and AF demagnetisation. Component I was isolated in the peridotites from the lower part of the sill (Table 1, entry 1). In most cases it seems to be a single component, common to all magnetic carriers in these samples [Fig. 5(a)] that have been established by the mineralogical analysis (see Section 4). However component II (see below) was also isolated in two such samples at temperatures below 580°C [see the example in Fig. 5(e)]. Here it is probably carried by secondary magnetite. Additionally, component I has been isolated in baked Suisarian schists near its contact with the sill (Table 1, entry 2), usually after treatment at high temperatures or high alternating fields [Fig. 5(b)]. Unfortunately, at temperatures higher than 550°C, mineralogical changes occur in the schists, indicated by a sharp increase in magnetic susceptibility and in the noisy behaviour of the remanence. Consequently, the stable end point is often unachievable for these rocks and statistics are poor. A stereoplot of sample mean directions for the component I is shown in Fig. 6(a). Statistics are presented in Table 1 (entries 1, 2 and 7).

Component II has been obtained in Suisarian picritic basalts and schists far from the sill [site C, Fig. 1(b)]. This component was mainly isolated at temperatures of 200–450°C, sometimes at higher temperatures. It is a single component for about one half of these samples [Fig. 5(c)]. However, component III (see below) was also isolated in six samples of these rocks at lower temperatures. In these cases, components II and III were separated at about 400°C [Fig. 5(f)], but this temperature slightly varies from sample to sample. Some samples of peridotite from lower and central parts of the sill possess this component with similar or slightly higher blocking temperatures [Table 1, entry 3, Fig. 5(e)]. Fig. 6(b) shows the stereoplot of sample mean directions for this component.

Component III [Fig. 5(d)] has mainly been isolated in peridotites and picrites from the upper

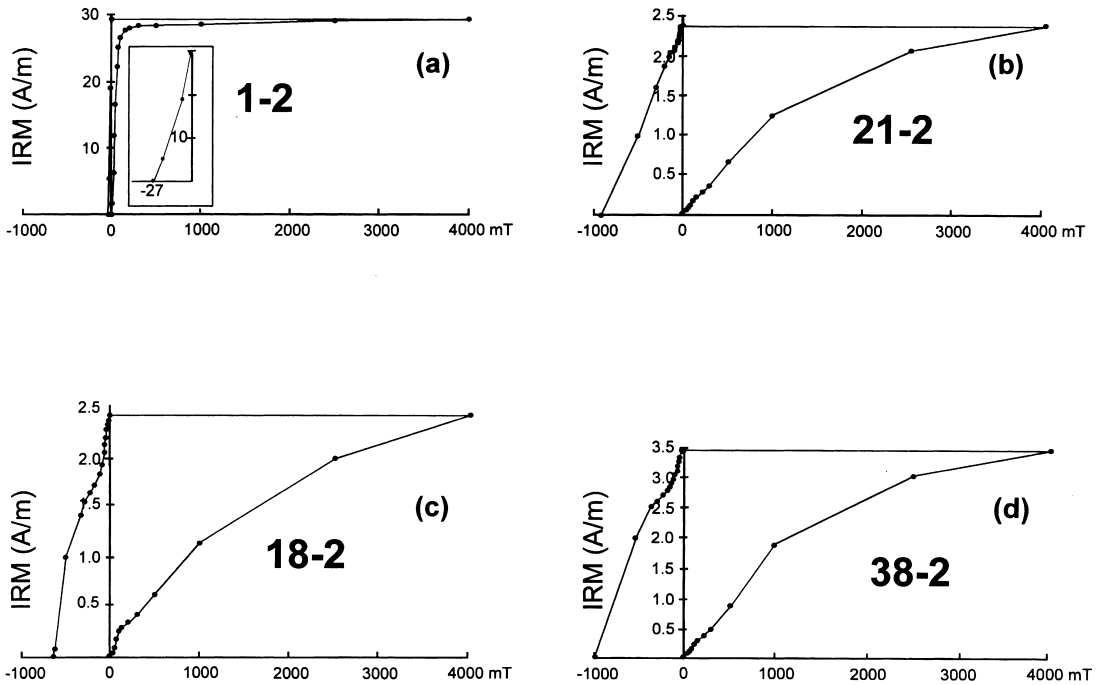


Fig. 4. Examples of IRM saturation curves: (a) peridotite, bottom part of the sill (site A); (b) peridotite, central part of the sill (site B); (c) baked schist (site B); (d) picritic basalt (site C).

and central parts of the sill (sites A and B) where it is a single stable component. As it was mentioned, this component was also isolated in some picritic basalts and schists at relatively low temperatures during thermal demagnetisation (Table 1, entry 6). A stereoplot of sample mean directions for the component III is shown in Fig. 6(c).

Most of the studied rocks also possess a less-stable present-day component of NRM [Fig. 6(d)].

We did not find any outcrops on the opposite limb of the studied structure [Fig. 1(b)], so it was impossible to carry out a fold test, and the conglomerate test was also impossible due to the absence of convenient rocks. The only field test available was the baked contact test (e.g. Piper, 1987).

Fig. 7 shows that the palaeomagnetic directions of peridotites from the lower part of the sill (site A) are similar to those of the baked schists (component I). This component was not isolated in the central or upper parts of the sill (sites A and B), or in the country rocks far from the contact (site C). From the peridotites and picrites in the other parts of the sill and from the picritic basalts and schists far from the contact, only components II and III were isolated. These results, together with rock magnetic tests, give some indirect evidence of the primary origin of component I. We suggest the following explanation. Peridotites in the bottom part of the sill probably possess a primary magnetisation (component I), having formed during the cooling of the body. The magnetisation of schists at the baked contact has also survived since the time of intrusion. As for the other parts of the sill, country picritic basalts and schists, they were probably remagnetised later during the Svecofennian tectonic epoch, and at least two remagnetising events occurred (components II and III). As was mentioned above (see Section 2), it is not quite clear whether the sill was intruded horizontally or some tilt already existed during emplacement. As it is supposed that component I is primary, the corresponding calculations were made both with (Table 1, entry 10) and without (Table 1, entry 7) tilt correction. For this we used the mean bed positions of the country rocks near

Table 1
Konchozero sill and country rocks, mean directions and palaeopoles (Lat = 62.1°N, Long. = 34.0°E)

Structure	Rock type	N/n	Decl. (°)	Incl. (°)	k	α_{95} (°)	P_{lat} (°N)	P_{long} (°E)	D_p (°)	D_m (°)	Component of NRM
1	Lower part of the Konchozero sill, peridotites, site A	6/9	105.6	42.9	21.1	14.9	−15.0	278.7	11.4	18.4	I
2	Baked schists, site B	5/9	100.1	36.4	13.4	21.7	−13.2	285.6	14.7	25.3	I
3	Lower and central parts of the Konchozero sill, peridotites, site A	8/8	347.6	41.1	48.5	8.0	50.6	232.0	5.9	9.7	II
4	Central and upper parts of the Konchozero sill, peridotites and picrites, sites A and B	12/15	13.7	39.4	108.5	4.2	49.1	194.4	3.0	5.0	III
5	Pictritic basalts and schists, high-T comp., sites B and C	12/17	347.5	40.0	34.5	7.5	49.7	232.0	5.4	9.0	II
6	Picritic basalts and schists, low-T comp., site C	6/6	23.6	43.3	73.3	7.9	50.0	179.8	6.1	9.8	III
7	Comp.I (1+2)	11/18	103.0	40.0	17.9	11.1	−14.2	281.9	8.0	13.4	I
8	Comp.II (3+5)	20/25	347.6	40.5	40.9	5.2	50.1	231.9	3.8	6.3	II
9	Comp.III (4+6)	18/24	16.8	40.8	80.1	3.9	49.6	189.8	2.9	4.7	III
10	Comp.I, with tilt correction	11/18	116.1	42.9	17.5	11.2	−10.7	270.0	8.6	13.9	I

N = number of samples from each site.
n = number of specimens.
Decl. and Incl. = site mean declination and inclination values, respectively.
k = best estimate of the precision parameter of Fisher (1953).
α_{95} = the semi-angle of the 95% cone of confidence.
P_{lat} = latitude of the calculated palaeopole.
P_{long} = the calculated longitude of the palaeopole.
D_p and D_m = the semi-axes of the cone of confidence about the pole at the 95% probability level. Mean directions are based on sample data.

sites A and B [Fig. 1(b)]. We used the untilted palaeomagnetic direction of component I in further discussion. However, the tilt correction does not change this direction significantly, as one can see from Table 1, so our main conclusions would be the same even in the case of pre-folding magnetisation.

6. Discussion

Precambrian palaeomagnetic data from the Fennoscandian Shield have variable time coverage of the geochronological scale. As a consequence, the reliability of the Fennoscandian APWP (Elming et al., 1993) varies. For example, there are many good results for Svecofennian and post-Svecofennian times (ca. 1880–1760 Ma). Hence the corresponding part of APWP seems to be quite reliable and it can be used for palaeomagnetic dating. The situation is, however, quite different for early Palaeoproterozoic times. The number of palaeomagnetic determinations is much less here, only a few were carried out on reliably dated rocks and tested for the primary nature of the ChRM. Such results are compiled in Table 2 together with our new results (component I).

Fig. 8 shows these palaeopoles together with the fragment of Fennoscandian APWP (Elming et al., 1993) for the 1700–2400 Ma period.

The Sm–Nd age of the Konchozero sill is 1974 ± 27 Ma (Puchtel et al., 1995); consequently, the age of any isolated component cannot be older. The results of the study of the baked contact, rock magnetic tests, together with the behaviour of NRM during demagnetisation suggest that component I corresponds to this age, or at least it is older than the other two components. However, this palaeopole is close to the group of poles with ages of 2436–2450 Ma. Mertanen et al. (1989)

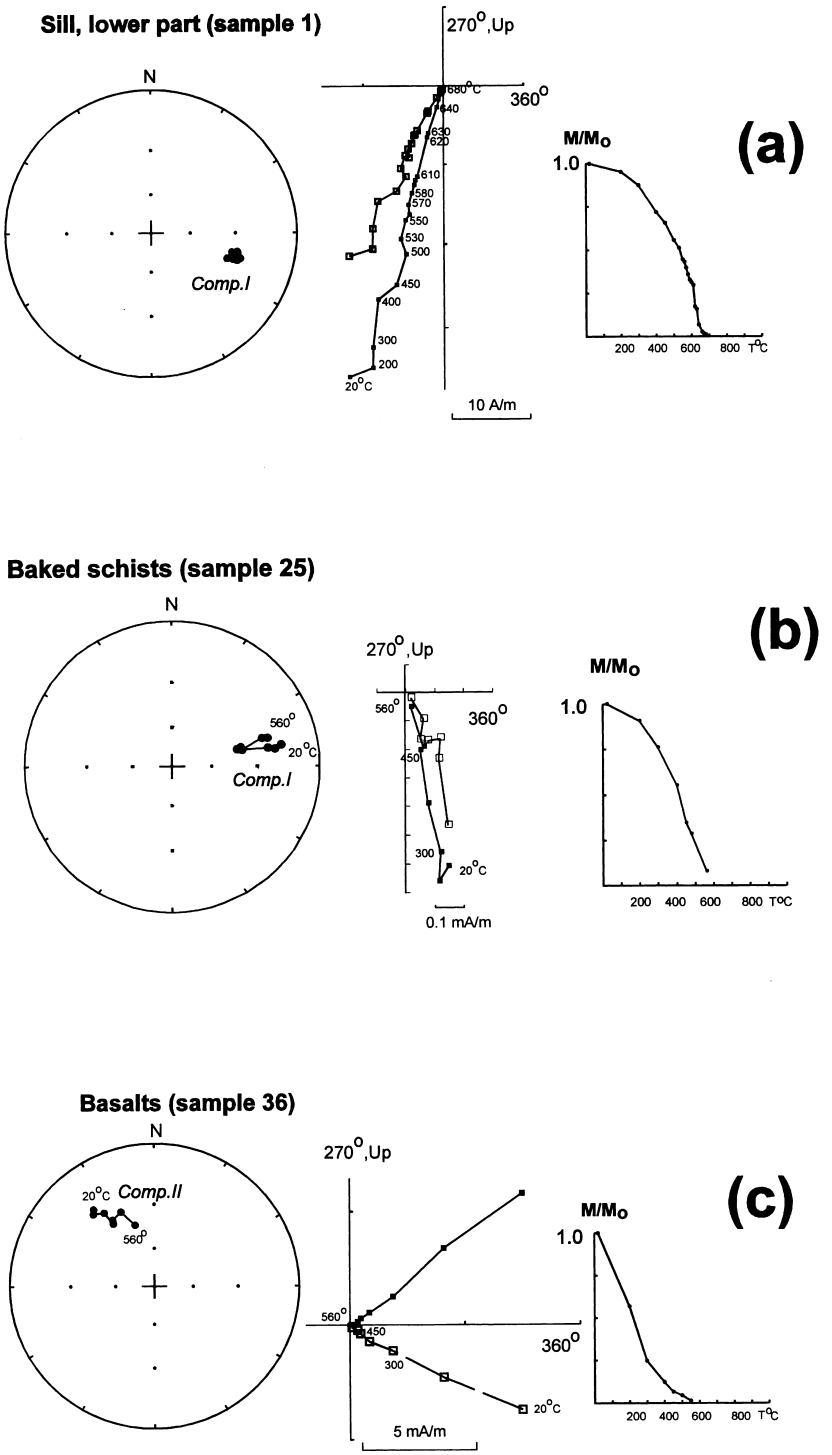
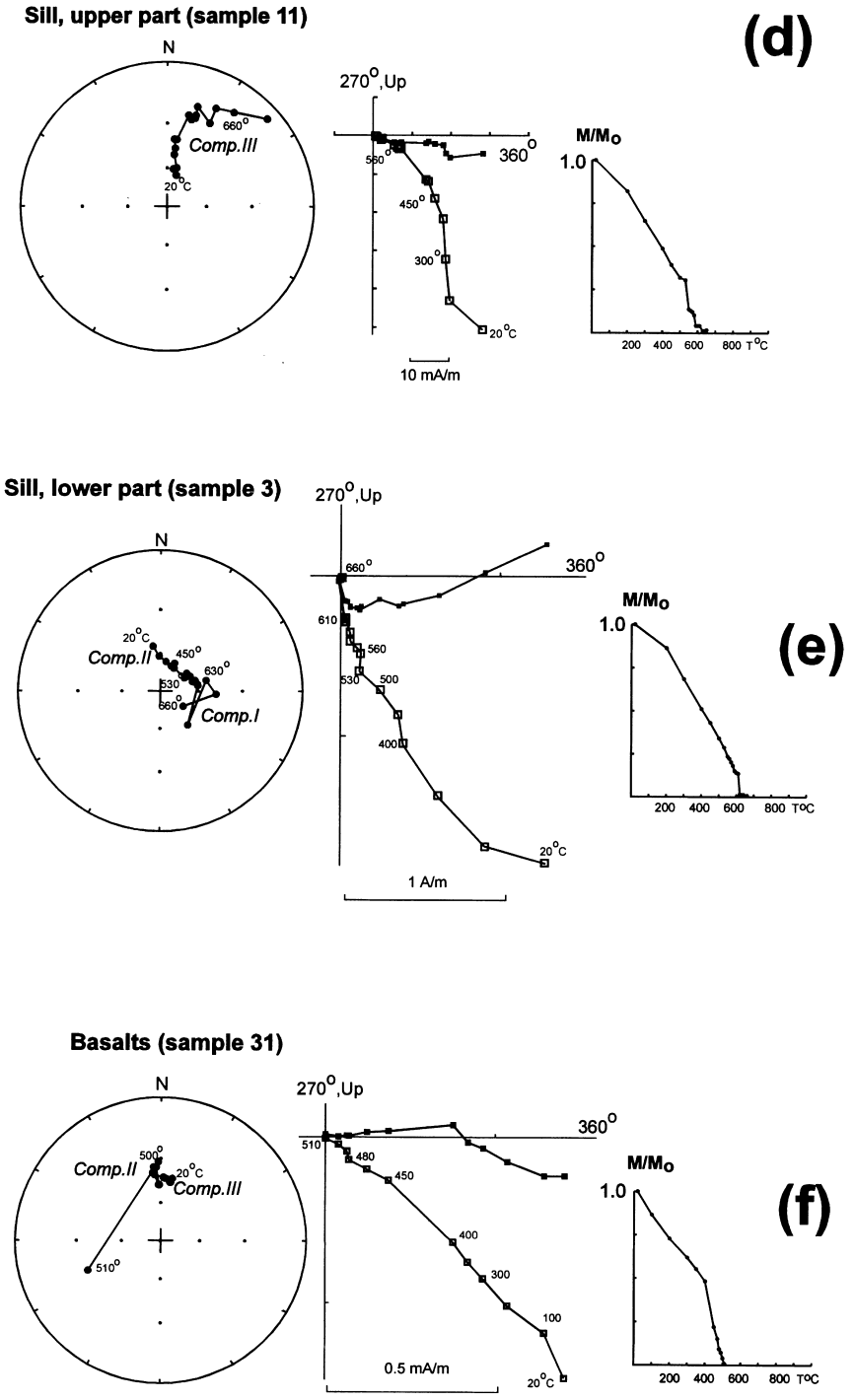


Fig. 5. Examples of thermal demagnetisation (equal angle projections, orthogonal plots and normalised demagnetisation curves) of: (a) the peridotite from the bottom part of the Konchozero sill (site A); (b) the baked schist on the contact of the Konchozero sill (site B); (c) the picritic basalt far from the contact (site C); (d) the peridotite from the top part of the Konchozero sill



(site A); (e) the peridotite from the bottom part of the Konchozero sill (site A), multi-component case; (f) the picritic basalt far from the contact (site C), multi-component case. Solid and open symbols on the stereoprojection represent downward and upward directions, respectively; solid and open symbols on the Zijderveld (1967) diagram denote horizontal and vertical projections, respectively.

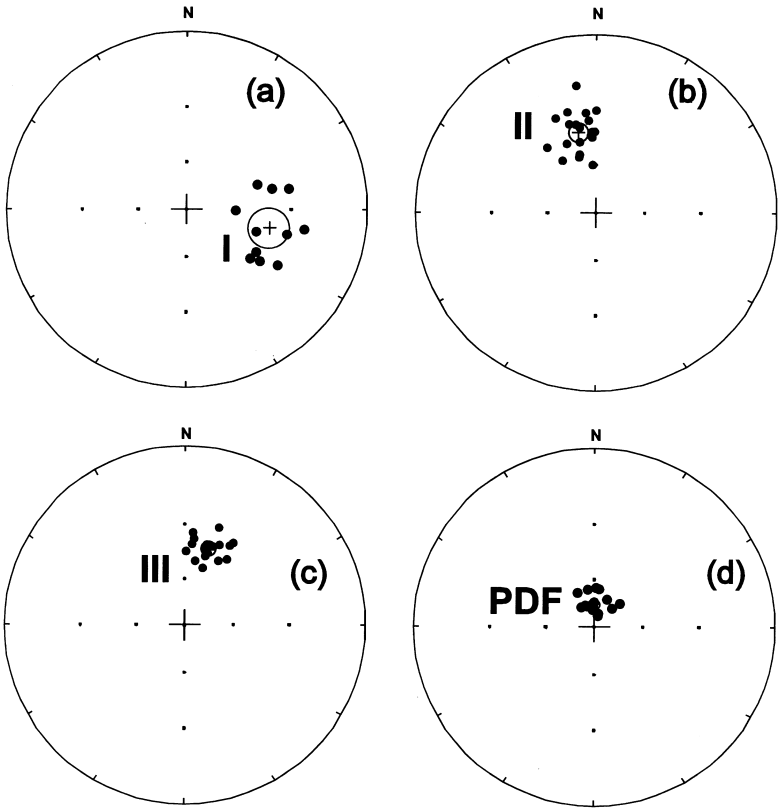


Fig. 6. Stereoplots (equal angle projection) of sample mean palaeomagnetic directions for: (a) component I; (b) component II; (c) component III; (d) present-day field component (PDF).

called these directions the D-component. This component was isolated both in the Karelian and the Kola provinces. Mertanen and Pesonen (1994) have also isolated the D-component in the Tsuomasvarri gabbro-diorite intrusion dated at 1930 Ma together with the A′-component (Table 2). Later, Mertanen (1995) said that the D-component in the Tsuomasvarri intrusions is probably a Vendian secondary overprint since the Tsuomasvarri area is adjacent to the areas of the Varanger and Kola peninsulas where dykes of that age occur. However the latitudes of palaeomagnetic poles obtained from Sredni and Komagnes Vendian dykes situated to the south of the Trollfjorden–Komagelva Fault Zone are 30–40° higher than those of D-poles (Torsvik et al., 1995, Table 2). So we think that the primary origin of the D-component in the Tsuomasvarri intrusion is still possible. There is some evidence in favour of the primary origin of the A′ component too, so we placed both of these components in Fig. 8. Mertanen and Pesonen also mentioned unpublished results of Elming who isolated this D-component in greenstones and metadiabases of northern Sweden with a maximum age of ~2.2 Ga.

The palaeopole of Torsvik and Meert (1995) from the Pechenga Zone is close to the APWP, but its age needs to be confirmed (these authors mention an age of 2125 Ma with reference to private communications; see note to Table 2).

There are four possible explanations for the close agreement of poles from such different age intervals.

(1) Pole positions for the 1900–2000 Ma and the 2430–2450 Ma intervals do not differ significantly due to extremely low speed of drift during this period.
(2) This concordance is due to a loop on the

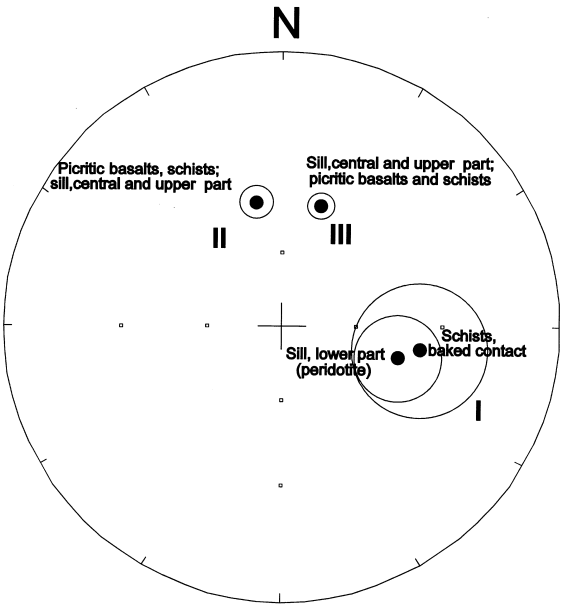


Fig. 7. The combined results of different rock types.

APWP covering this period. There are several cases of such behaviour; for example, poles from Fennoscandian APWP (Elming et al., 1993) for ca. 1700 and ca. 1310 Ma are very close to each other.

(3) This magnetisation corresponds to Ludikovian age, and rocks with ages of ca. 2440 Ma were remagnetised at 1900–2000 Ma.
(4) All these rocks were remagnetised later, in Vendian time; for example Torsvik and Meert (1995) mentioned such a possibility for the low-temperature B-component isolated in rocks of the Pechenga Zone.

New palaeomagnetic data from rocks of ages between 1900 and 2500 Ma are required to solve this problem.

Only in case of the fourth possibility would the direction of component I from the Konchozero sill not correspond to the age of the intrusion. However, this suggestion seems unlikely, because the Konchozero sill is far from the area of Vendian magmatic activity. The study of baked contact, together with the rock magnetic studies, points to primary origin for this magnetisation.

Components II and III are in good agreement with the well-determined part of the Fenno-scandian APWP (Elming et al., 1993) between 1860 and 1760 Ma. According to this curve, the age of component II is 1860 Ma and that of component III is 1760 Ma. There is much evidence of Svecofennian remagnetisation in the rocks of the Fennoscandian shield (Mertanen, 1995).

Table 2
Radiometrically dated early Proterozoic palaeomagnetic poles for the Fennoscandian shield

N	Unit	Age (Ma)	Method	S_{lat} (°N)	S_{long} (°E)	P_{lat} (°N)	P_{long} (°E)	D_p (°)	D_m (°)	Reference
1	Tsuomasvarri gabbro-diorite intrusions, D	1931±2	U–Pb	69.9	28.3	−19.7	285.2	13.8	22.9	Mertanen and Pesonen (1994)
2	Tsuomasvarri gabbro-diorite intrusions, A′	1931±2	U–Pb	69.9	28.3	40.2	247.3	4.7	7.7	Mertanen and Pesonen (1994)
3	Konchozero peridotite sill, Comp. I	1974±27	Sm–Nd	62.1	34.0	−14.2	281.9	8.0	13.4	This study
4	Kuetsyarvi formation, andesites and pillow basalts	2125[a]	?	69.5	29.6	24.7	300.8	13.0	21.4	Torsvik and Meert (1995)
5	Koillismaa layered intrusion, D	2436±5	U–Pb	65.8	28.6	−18.0	265.0	A_{95}=23		Mertanen et al. (1989)
6	Suopera subophitic dykes, D	2446±5	U–Pb	65.7	30.2	−31.0	277.1	10.8	14.9	Mertanen (1995)
7	Main range gabbro-dolerites	2436±25	U–Pb	68.0	32.5	−16.7	304.8	6.7	12.3	Khramov et al. (1997)
8	Burakovka layered intrusion	2449±1	U–Pb[b]							
		2439±20	Sm–Nd	62.1	36.2	−19.5	262.6	6.1	8.5	Khramov et al. (1997)

S_{lat}, S_{long} = latitude and longitude of the sampling area.
P_{lat}, P_{long} = latitude and longitude of the palaeopole.
D_p, D_m = the semi-axes of the cone of confidence about the pole at the 95% probability level.
[a]Age mentioned in Torsvik and Meert (1995) (with reference to private communications with Melezhik).
[b]Age from Amelin et al. (1995).

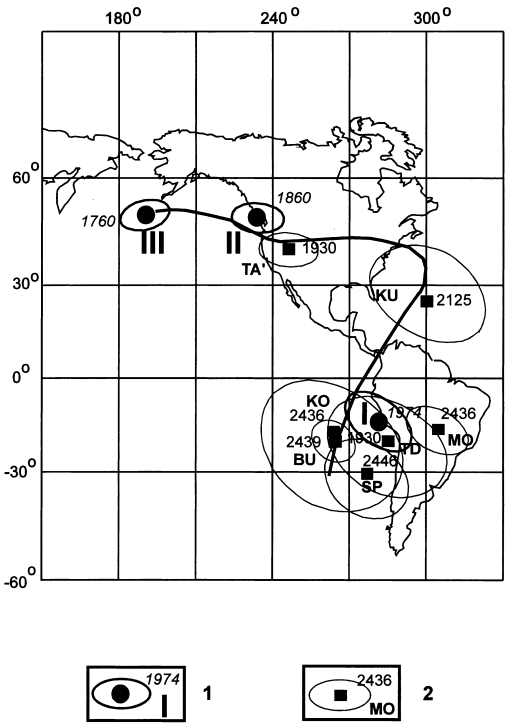


Fig. 8. The ca. 1800–2400 Ma part of the Fennoscandian APWP (Elming et al., 1993): 1, new palaeopoles from Konchozero rocks, roman figures present components of ChRM; 2, dated palaeopoles for ages 1900–2500 Ma; TA′, TD, Tsuomasvarri gabbro-diorite intrusion, components A′ and D (Mertanen and Pesonen, 1994); KU, Kuetsyarvi Formation (Torsvik and Meert, 1995); BU, MO, Burakovka intrusion and Monchegorsk pluton (Khramov et al., 1997); SP, Suopera and Paajarvi dykes (Mertanen, 1995), see Table 2.

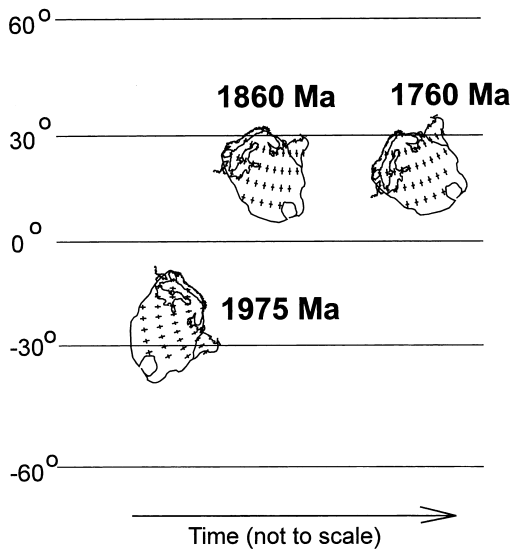


Fig. 9. Three palaeolatitudal and azimuthal positions of Baltica based on the results from the Konchozero sill (see Table 1).

Fig. 9 shows three proposed palaeolatitudal and azimuthal positions of Baltica (we use this term for the tectonic unit including the Fennoscandian and Ukrainian shields and Volga-Urals tectonic unit; according to Bogdanova et al. (1994) they have been connected since ca. 2000 Ma) based on the results of this study.

7. Conclusions

Three components of NRM have been isolated during thermal and AF demagnetisations of peridotites and picrites of the Konchozero sill, baked schists and country picritic basalts of Ludikovian age (ca. 2100–1950 Ma).

We suggest that our pole position for component I corresponds to the Sm–Nd age of 1974 ± 27 Ma Puchtel et al., 1995.

Pole positions for components II and III are consistent with the well determined part of the APWP of Elming et al. (1993) between 1880 and 1760 Ma (Fig. 8). These two components are probably the result of two different remagnetisation events during the Svecofennian tectonic epoch. Corresponding ages of these events are ca. 1860 and 1760 Ma.

The closeness of our component I pole with poles from rocks with the age of ca. 2440 Ma can be explained either by tectonic reasons such as slow plate movement or an APWP loop, or by remagnetisation. New palaeomagnetic data from the time interval between ca. 2500 and ca. 1900 Ma are necessary to resolve this problem.

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References

Amelin, Yu.V., Heaman, L.M., Semenov, V.S., 1995. U-Pb geochronology of layered mafic intrusions of the eastern Baltic Shield: implications for the timing and duration of Paleoproterozoic continental rifting. Precamb. Res. 75, 31–46.

Bogdanova, S.V., Bibikova, E.V., Gorbachev, R., 1994. Palaeoproterozoic U–Pb zircon ages from Belorussia: new geodynamic implications for the East European craton. Precamb. Res. 68, 231–240.

Briden, J.C., Duff, B.A., 1981. Pre-Carboniferous palaeomagnetism of Europe north of the Alpine orogenic belt. In: McElhinny, M.W., Valencio, D.A. (Eds.), Paleoreconstructions of the Continents. Am. Geophys. Union., Geodynam. Ser. 2, pp. 137–150

Elming, S.-A., Pesonen, L.J., Leino, M.A.H., Khramov, A.N., Mikhailova, N.P., Krasnova, A.F., Mertanen, S., Bylund, G., Terho, M., 1993. The drift of the Fennoscandian and Ukrainian shields during the Precambrian: a palaeomagnetic analysis. Tectonophysics 223, 177–198.

Fisher, R.A., 1953. Dispersion on a sphere. Proc. R. Soc. Lond. A 217, 295–305.

Gooskova, E.G., Krasnova, A.F., 1985. Palaeomagnetism of the Archean and Proterozoic basic intrusions of the eastern part of Baltic shield. Fizika Zemli N5, 58–68 (in Russian).

Khramov, A.N., Fedotova, M.A., Pisakin, B.N., Priyatkin, A.A., 1997. Palaeomagnetism of Lower Proterozoic intrusions and accociated rocks of Karelia and Kola peninsula: the contribution and elaboration of the model of the Pre-Cambrian evolution of Russian-Baltica craton. Fizika Zemli N6, 24–41 (in Russian).

Kulikov, V.S., Rybakov, S.I., Golubev, A.I., Svetov, A.P. (Eds), 1987. Guide of geological excursions in Karelia. Petrozavodsk, KFAN SSSR, p. 94.

Mertanen, S., 1995. Multicomponent remanent magnetizations reflecting the geological evolution of the Fennoscandian shield — a palaeomagnetic study with emphasis on the Svecofennian orogeny. Geological Survey of Finland, Espoo, p. 46.

Mertanen, S., Pesonen, L.J., 1994. Preliminary results of a palaeomagnetic and rock magnetic study of the Proterozoic Tsuomasvarri intrusions, northern Fennoscandia. Precam. Res. 69, 25–50.

Mertanen, S., Pesonen, L.J., Huhma, H., Leino, M.A.H., 1989. Palaeomagnetism of the early Proterozoic layered intrusions, Northern Finland. Geological Survey of Finland 347, p. 40.

Piper, J.D.A., 1987. Palaeomagnetism and the continental crust. Open Univ Press, Milton Keynes, New York-Toronto, p. 434.

Puchtel, I.S., Bogatikov, O.A., Kulikov, V.V., Kulikova, V.V., Zhuravlev, D.Z., 1995. The role of crust and mantel sources in the petrogenesis of continental magmatism: isotopic and geochemical data from Early Protaerozoic picrites of Onega plateau, Baltic shield. Petrologia 3, N4, 397–419 (in Russian).

Sokolov, V.A., Kulikov, V.S., Stenar, M.M. (Eds.), 1987. Geology of Karelia. Nauka, Leningrad, p. 231 (in Russian).

Torsvik, T.H., 1986. Interactive analysis of palaeomagnetic data, IAPD user guide. University of Bergen, pp. 1–74.

Torsvik, T.H., Meert, J.G., 1995. Early Proterozoic palaeomagnetic data from the Pechenga Zone (north-west Russia) and their bearing on early Proterozoic palaeogeography. Geophys. J. Int. 122, 520–536.

Torsvik, T.H., Roberts, D., Siedlecka, A., 1995. Palaeomagnetic data from sedimentary rocks and dolerite dykes, Kildin island, Rybachi, Sredni and Varanger Peninsulas, NW Russia and NE Norway: a review. Norweg. Geolog. Bull. (special publication) 7, 1–12.

Zijderveld, J.D.A., 1967. A.C. demagnetization of rocks: analysis of results. In: Collinson, D.W., Creer, K.M., Runcorn, S.K. (Eds.), Methods in Palaeomagnetism, Elsevier, Amsterdam, pp. 254–286.

Zonenshain, L.P., Kuzmin, M.I., Natapov, L.M., 1990. Tectonics of the lithospheric plates of the territory of USSR. Nedra, Moscow 1, 328(in Russian)